AIRTEL SYNC HACKATHON 2026 • ENTERPRISE TRACK A

Project Udyam Digital: Data Privacy & Security Charter

Architectural Design Choices for DPDP Act 2023 & Telecom Regulatory Compliance

Classification: Hackathon Submission Pack (1-Page Brief)
Statutory Basis: DPDP Act 2023 & DoT Unified License

✓ Privacy-by-Design Verified

Executive Design Rationale: Project Udyam Digital re-engineers the legacy 17.6-day SMB acquisition cycle into an automated, self-service digital onboarding journey. Because telecom Customer Application Forms (CAF) process sensitive business identities, the architecture is founded on *Privacy-by-Design and Default* principles compliant with the **Digital Personal Data Protection Act, 2023 (DPDP Act)** and **DoT License Conditions**. Collection is strictly minimized, consent is explicit and unbundled, PII is masked across all UI views and observability logs, and data retention is governed by statutory lifecycles.

1. ADVISOR CHAT — High-level profiling only; zero PII ingested → **2. GRANULAR CONSENT** — Unbundled affirmative opt-in with SHA-256 hash → **3. SMART KYC LOOKUP** — Ephemeral mock registry queries; zero doc hoarding → **4. TOKENIZED STORAGE** — AES-256-GCM + Salted masks in UI & traces → **5. STATUTORY PURGE** — 5-yr DoT retention, then auto-zero-fill purge

1 Purpose Limitation & Minimization (DPDP §5)

Traditional enterprise telecom sales demand paper photocopies of bank statements, utility bills, and personal identity cards. Udyam Digital eliminates data hoarding:

- **Stateless Registry Lookups:** Verification connects to national GSTIN (3,534 mock records) and Udyam (2,723 mock records) registries via stateless API lookups. Only the verified status, entity trade name, and validation token are kept; raw query payloads are immediately discarded.
- **No Banking Credential Exposure:** Recurring auto-pay mandates use tokenized NPCI / e-NACH handles. Zero bank account numbers or debit cards are stored on platform servers.
- **Static Postal Geometry Only:** Feasibility calculation relies strictly on user-supplied premises PIN codes and shop numbers. Persistent GPS or mobile device network tracking is strictly prohibited.
- **Zero Data Monetization:** Enterprise records are ring-fenced for connectivity provisioning; cross-selling to 3rd-party advertisers or data brokers is programmatically barred.

2 Granular Consent & Verifiable Audit (DPDP §6)

Per Section 6 of the DPDP Act 2023, consent must be **free, specific, informed, unconditional, and unambiguous** with verifiable proof:

- **Affirmative Pre-KYC Checkpoint:** Before uploading documents or triggering automated registry lookup, users encounter an unbundled consent screen detailing purpose, data items collected, and retention boundaries. No hidden terms or pre-ticked boxes.
- **Cryptographic SHA-256 Proof:** Each consent submission generates an immutable digital token: `{timestamp_utc, principal_id, ip_hash, consent_version, purpose}` hashed with SHA-256 for non-repudiation during regulatory audits.
- **Unbundled Channels:** Statutory e-KYC consent is completely decoupled from optional promotional communications, WhatsApp alerts, and partner SaaS offers.
- **Unilateral Right to Revoke:** Signatories can retract secondary marketing consent anytime via the self-service Order Tracker without interrupting primary telecom services.

3 Dynamic Identifier Masking & Zero-Trust RBAC

To prevent accidental PII leakage across browser storage, partner portals, and telemetry logging pipelines, all direct identifiers are dynamically salted and masked:

IDENTIFIER	RAW INPUT FORMAT	UI PRESENTATION MASK	TELEMETRY / LOG TRACE
GSTIN	`27AABCA1234F1Z5`	`27AAB*****1Z5`	`GST_HASH_8F92`
Aadhaar VID	`9182 3847 1928 4719`	`XXXX-XXXX-4719`	`[REDACTED_UIDAI_TOKEN]`
Mobile No.	`+91 98104 58210`	`+91 9810X-XXXX`	`TEL_MASK_981XX`
PAN Card	`AABCA1234F`	`AAB****34F`	`PAN_TOKEN_E91A`
Udyam No.	`UDYAM-MH-00-0012345`	`UDYAM-MH-**-****345`	`UDYAM_TOKEN_4F2C`

Zero Plaintext Storage: Field reps and support agents see masked records only. Sensitive unmasking requires dual-custody authorization keys inside secure enclaves.

4 Statutory Retention vs. Erasure (DPDP §8(7) & DoT)

While DPDP Act Section 8 requires prompt erasure, Section 8(7) protects sectoral statutory compliance. Airtel reconciles privacy with telecom licensing mandates:

- **Mandatory 5-Year CAF Retention:** Under **DoT Unified Access License Condition 39.20**, customer verification files (CAF, e-KYC records, and authorized signatory tokens) are legally mandated to be retained for exactly 5 years following service disconnection.
- **2-Year Call Detail Records (CDR):** Airtel IQ Cloud PBX and VoIP logs are retained for 2 years for statutory security audits, stored in write-once audit storage.
- **Automated Cryptographic Purge:** When the statutory retention period lapses, an automated zero-fill purge engine permanently destroys archived records and issues a deletion receipt.
- **Dropped Lead Sanitization:** Incomplete lead applications abandoned during early assessment are permanently wiped from sales caches after 90 days.

🔒 Transit & Rest Cryptography

TLS 1.3 Strict: Enforced ECDHE forward secrecy and HSTS for all web/API traffic.
AES-256-GCM: Authenticated volume & database encryption with automated 90-day HSM key rotation.

IN Sovereign Cloud Hosting

100% Domestic Residency: Hosted on Tier-IV *Nxtra by Airtel* sovereign data centers (Mumbai, Pune, Manesar). Zero cross-border data transfer outside the Union of India.

🛡 Data Principal Rights

Section 11-13 Rights: Self-service portal tools for Right to Access summary, Right to Correction/Update, Right to Grievance Redressal (15-day SLA), and Signatory Nomination.

✓ **Statutory Assurance:** Designed in alignment with the Data Protection Board of India (DPBI) rules and CERT-In cybersecurity directives.

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